

Introduction to Oracle Linux

Objectives

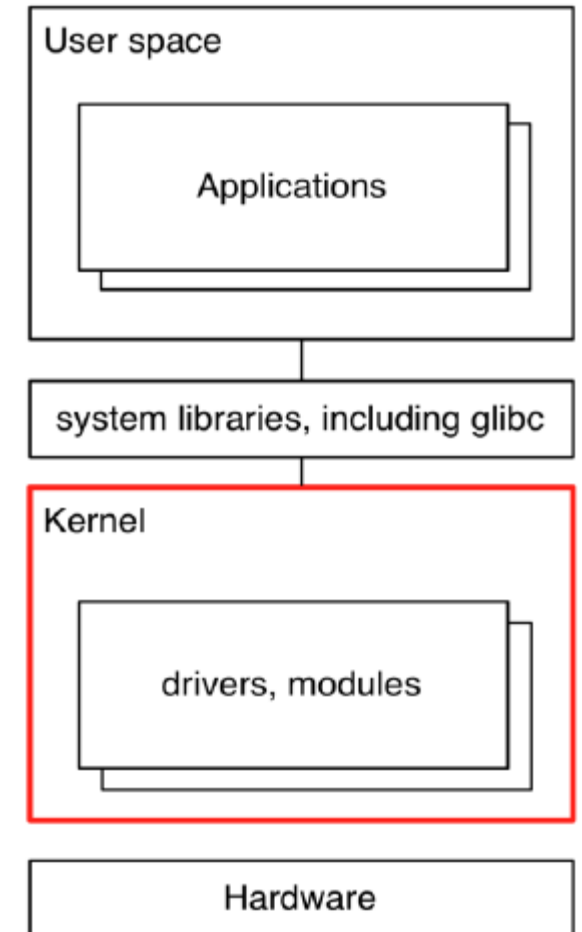
After completing this lesson, you should be able to describe:

- The history of the Linux operating system
- The Linux kernel development model
- Linux distributions
- Oracle's comprehensive Linux solution
- Oracle's contributions to the Linux community
- Oracle Linux's compatibility with Red Hat Enterprise Linux (RHEL)
- The Unbreakable Enterprise Kernel
- Dtrace and Btrfs



Linux Kernel

- Linux is modular in design:
 - User space
 - Kernel
- Modular design allows for a large development community, better fault isolation, and security.
- Linus Torvalds developed the original Linux kernel.
- Linux version 0.01 was released in September 1991.
- The name Linux is a combination of Linus and UNIX.



The GNU Project

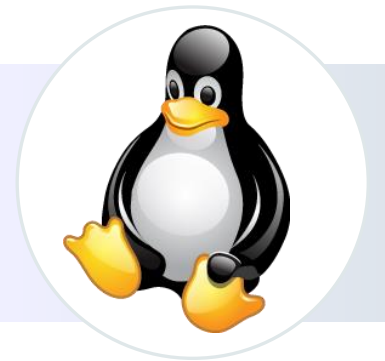
- The GNU Project was launched in 1983 by Richard Stallman.
- The goal was to create a free, UNIX-compatible operating system.
- GNU stands for “GNUs Not UNIX.”
- The GNU Project created many programs but no kernel.
- The Linux kernel filled the last gap in the GNU system.

GNU General Public License (GPL)

- Richard Stallman wrote the GPL for the GNU Project.
- GPL provides basic software freedoms:
 - Freedom to copy, change, and redistribute software
- Distributors must provide the source code at no cost.
- Linux kernel version 0.12 was licensed under the GNU GPL.
- The Linux community participates in the advancement of Linux.
- Other free software licenses exist, under which different Linux software packages are licensed.

Linux Kernel Development Model

- Thousands of developers contribute to frequent releases of the kernel.
- Features are pushed upstream through mail lists and IRC.
- New releases deliver stable updates, new features, and performance improvements.
- The Linus Torvalds–led team makes the new releases.
- Mainline kernels are released approximately every two to three months.
- Kernel branches are available at <http://www.kernel.org>.
- Linux kernel development uses Git as the source-code control system.



Continuous Mainline Kernel Development

- New hardware brings massive scalability changes and challenges.
 - High Input/Output Operations Per Second (IOP/s) in networking and storage
 - Dramatic bottlenecks in large symmetric multiprocessing (SMP) systems
- Performance is very dependent on power management.
- Mainline kernels have changes to address performance on new hardware.
- Oracle Linux 7 is supported on x86-64 (64 bit) hardware architectures.

Linux Distributions

- Linux distributions:
 - Are built on top of the Linux kernel
 - Are complete operating systems and more
 - Include compiled binaries and source code
- There are hundreds of Linux distributions:
 - Commercially backed distributions
 - Linux community–driven distributions
- Examples:
 - Oracle Linux, Debian, Fedora, Red Hat Enterprise Linux (RHEL), Ubuntu, and many others

Oracle Linux

- Brings the latest Linux innovations to customers
- Is the best-performing, most modern and reliable Linux OS
- Tracks mainline closely
- Influences Linux roadmap upstream via direct code contributions
- Provides highest-value, enterprise-class support
- Deployment best practices: Full stack tested with real-world workloads
- Provides comprehensive legal indemnification
- Lowers cost
- Oracle Ksplice: Apply kernel and select user space library updates on a running system



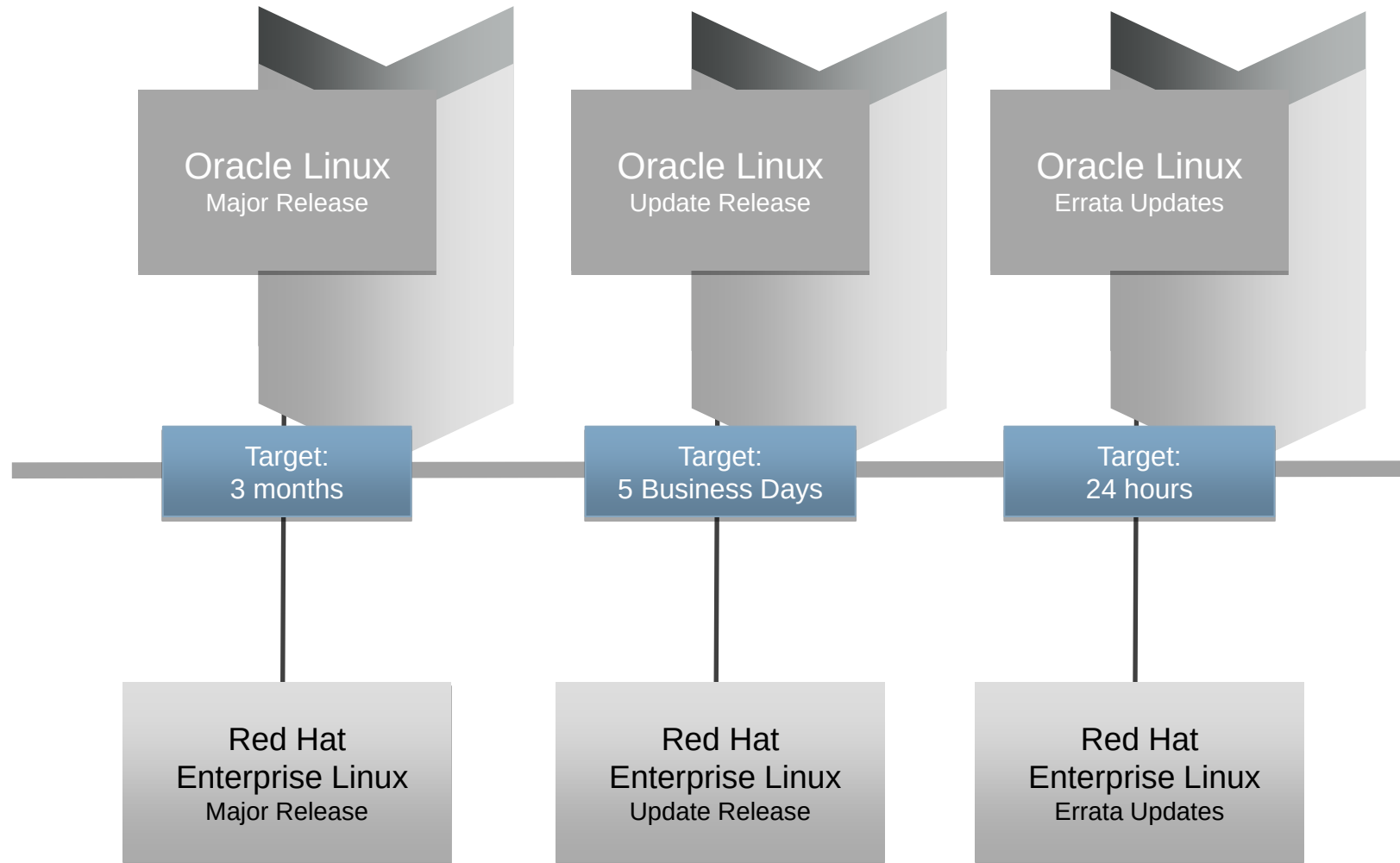
Oracle's Technical Contributions to Linux

- Oracle has a dedicated Linux kernel development team.
- Oracle's technical contributions to Linux include:
 - ASMLib
 - Asynchronous IO (AIO) Kernel Subsystem
 - Btrfs file system
 - Oracle Cluster File System (OCFS2)
 - Linux data integrity based on the T10-PI standard
 - Xen Hypervisor
- All Oracle Linux code is available to the Linux community.
- The Git source tree with change logs and commit messages is available at:
 - <http://oss.oracle.com/git/>

Oracle Linux: Compatible with Red Hat Enterprise Linux (RHEL)

- Source and binaries are fully compatible with RHEL.
- Applications that run on RHEL run on Oracle Linux.
- Trademarks and logos have been removed, but there are no compatibility issues.
- `/etc/oracle-release` was added to identify code obtained from Oracle.
- Oracle continues to track RHEL releases with Oracle Linux ISO releases and errata stream.

Oracle Linux Release/Update Targets Relative to Red Hat Enterprise Linux (RHEL) Releases/Updates



The Unbreakable Enterprise Kernel (UEK)

- Oracle announced the Unbreakable Enterprise Kernel in September 2010.
- It is used by Exadata and Exalogic for extreme performance.
- The Unbreakable Enterprise Kernel is available since Oracle Linux version 5 Update 5.
- Since Oracle Linux 5 Update 5, you have a choice:
 - Red Hat Compatible Kernel
 - The Unbreakable Enterprise Kernel
- Oracle is committed to offering compatibility with Red Hat.
- Full support is offered for customers running either kernel.
- Existing applications run unchanged.

DTrace

- DTrace:
 - Began as a Solaris tool, available since 2005
 - Is now fully integrated with Oracle Linux and UEK
 - Allows static tracing by using instrumentation compiled into kernel and applications
 - Allows dynamic tracing by defining probe points dynamically
- Probes and actions at probe points are defined by scripts written in the “D” language.
- Many types of providers:
 - DTrace, syscall, profile, sysinfo, vminfo, fpuinfo, sched, io, iscsi, and so on
 - A Pid provider for dynamic tracing in user space applications
- Speculative tracing allows the filtering of events and data presented to the user after probes fire.
- DTrace is available only with Premier ULN Support from Oracle.

Btrfs File System

- Designed for large files and file systems
- Simplified administration
- No volume manager needed
- Easy to add and remove capacity
- Online defragmentation and scrubbing
- Built-in data integrity
- RAID
- Flexible
- File and file subvolume snapshots
- Transparent compression



Oracle Linux Release Notes

- Oracle publishes release notes for each version.
- Release notes and all product documentation for Oracle Linux 7 are available at:
http://docs.oracle.com/cd/E52668_01/.
- New features and changes
- Known issues
- Installation and availability
 - Upgrading from Oracle Linux 6
- Changes from the Upstream Release:
 - Removed, modified, new packages from upstream release
 - Packages added by Oracle



Quiz



Oracle Linux offers a Red Hat-compatible kernel as well as a kernel that is optimized for Oracle applications.

- a. True
- b. False



Quiz



Which of the following statements are true about the Unbreakable Enterprise Kernel (UEK)?

- a. The UEK is the default kernel since Oracle Linux 5 Update 6.
- b. The UEK is heavily tested with Oracle workloads.
- c. Existing applications run unchanged when using the UEK.
- d. The UEK is the only kernel available with Oracle Linux.



Summary

In this lesson, you should have learned how to describe:

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Practice 2: Overview

This practice covers the following topics:

- Quiz – Introduction to Oracle Linux
- Viewing kernel information